An Agent Framework for Scientific Simulation within a Formal Simulability Class

Chengshuai Yang

NextGen PlatformAI C Corp, USA

Correspondence: integrityyang@gmail.com

March 13, 2026

Summary. Scientists in every discipline build bespoke computational solvers—finite-element codes for bridges, quantum-chemistry packages for molecules, epidemiological simulators for disease spread. Each solver works for one class of problems but cannot be reused for another, creating an expertise bottleneck that limits who can simulate what. Here we ask: is there a single framework that can simulate *any* problem within a rigorously defined class, with guaranteed error bounds? We identify such a class—the *simulability class* $\mathfrak{S}$, comprising all finitely specifiable, numerically approximable problems with explicit observables—and prove its sharp boundary against four fundamental mathematical obstructions. We show that structured specifications (`spec.md` files), validated and executed by a three-agent pipeline over 12 computational primitives, achieve bounded-error simulation for every problem in $\mathfrak{S}$. We validate this across 12 scientific domains including a real clinical CT dataset, demonstrate through ablation that each pipeline component is necessary, and show that every problem in $\mathfrak{S}$ is automatically benchmarkable—enabling open competition across all of simulable science.

Keywords: scientific event horizon, simulability class, specification-driven simulation, computational primitives, agent pipeline

1Introduction

Every year, thousands of scientists independently build numerical solvers. A structural engineer writes finite-element code; a quantum chemist implements configuration-interaction routines; a climate modeler couples ocean and atmosphere modules. Each solver is correct for its domain but cannot be reused for any other. This *expertise bottleneck*—the requirement that every new problem demands a new, hand-crafted computational pipeline—limits who can simulate what and creates a reproducibility crisis across computational science [16].

The mathematical foundations for unification exist. The Church–Turing thesis [1] guarantees that computable functions can be evaluated by a universal machine. The Lax equivalence theorem [2] ensures that consistent, stable discretizations converge. Hadamard's well-posedness theory [3] certifies continuous dependence on data. What has been missing is a framework that connects these classical results into a practical, automated pipeline.

Here we present such a framework. Our contributions are:

1. **The scientific event horizon**: a sharp boundary separating a rigorously defined simulability class $\mathfrak{S}$ from four fundamental obstructions that defeat *any* computational system (Proposition 2).
2. **Bounded-error realizability**: for any problem in $\mathfrak{S}$ and any tolerance $\varepsilon > 0$, the framework produces $\hat{x}$ with $\|\hat{x} - x^*\| \leq \varepsilon$ in finite time (Theorem 3).
3. **Specification-driven pipeline**: structured specifications (`spec.md`), not code, are the unit of scientific input. Three agents—Plan, Validation, Computation—transform specifications into solutions.

4. **Empirical validation**: 12 domains, one real clinical dataset, and ablation studies proving each pipeline component is necessary.

This extends our previous result on imaging-specific primitives [10] to the broader simulability class.

2 Results

2.1 The Simulability Class and Its Boundary

We begin by defining exactly what the framework covers and what it excludes.

Definition 1 (Simulability Class $\mathfrak{S}$)**.** *A scientific problem P belongs to the simulability class $\mathfrak{S}$ if and only if it satisfies all six conditions:*

1. ***Finitely specifiable**: P can be represented as a finite tuple $(\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$ where $\Omega \subseteq \mathbb{R}^d$ is the spatial domain, $\mathcal{E}$ are governing equations, $\mathcal{B}$ are boundary conditions, $\mathcal{I}$ are initial conditions, $\mathcal{O}$ is a set of observable quantities, and $\varepsilon > 0$ is the target tolerance.*
2. ***Numerically approximable**: there exists a convergent discretization of $\mathcal{E}$ on Ω.*
3. ***Expressible over the primitive basis**: the discretized system decomposes into a finite directed acyclic graph (DAG) over 12 computational primitives (Table 2).*
4. ***Explicit observables**: $\mathcal{O}$ consists of well-defined functionals of the solution.*
5. ***Explicit error metric**: a computable norm or metric quantifies the distance between computed and true solutions.*
6. ***Bounded resource assumptions**: the required computational resources (time, memory) are finite for any fixed $\varepsilon > 0$.*

This class is broad—it contains all well-posed PDEs, ODEs, integral equations, optimization problems, stochastic systems, and their compositions across physics, chemistry, biology, engineering, and applied mathematics. But it is not everything.

Proposition 2 (The Scientific Event Horizon)**.** *The complement $\mathfrak{U} = \mathfrak{S}^c$ is characterized by four fundamental obstructions, each of which defeats any conceivable computational system—classical, quantum, or hypothetical:*

1. ***Logical undecidability** (Gödel–Turing): problems reducible to the halting problem or Hilbert's tenth problem over $\mathbb{Z}$.*
2. ***Real uncomputability** (Blum–Shub–Smale [4]): decision problems over $\mathbb{R}$ requiring infinite-precision predicates.*
3. ***Quantum measurement irreducibility**: individual measurement outcomes governed by the Born rule (only distributions are computable).*
4. ***Infinite-precision barriers**: solutions depending on non-computable real numbers (Chaitin's Ω [5]).*

The boundary $\partial\mathfrak{S}$ is what we call the *scientific event horizon*. Unlike engineering limitations that yield to better hardware or algorithms, these four obstructions are theorems of mathematics: the halting problem will never be solved, the Born rule will never be circumvented, and Chaitin's Ω will never be computed. The event horizon is therefore a *fundamental constant of computational epistemology*—it delimits what any intelligence, human or artificial, can know through simulation, in the same way that the speed of light delimits what can be observed through causal contact. This is not a feature of our framework; it is a discovery about the structure of scientific knowledge itself. Our framework's contribution is to reach this boundary constructively from below. Table 1 catalogs representative problems at each stratum.

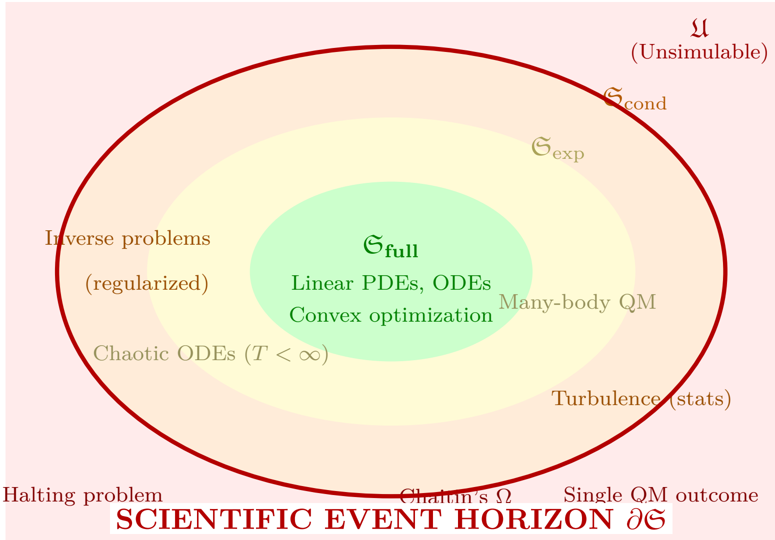

Figure 1. The simulability landscape. The event horizon $\partial\mathfrak{S}$ (bold red) separates the simulability class—problems solvable to arbitrary accuracy—from the unsimulable class $\mathfrak{U}$. Within $\mathfrak{S}$, three cost strata reflect the computational price of simulation: polynomial ($\mathfrak{S}_{\text{full}}$), exponential ($\mathfrak{S}_{\text{exp}}$), and conditional ($\mathfrak{S}_{\text{cond}}$, requiring regularization or statistical reformulation). Our framework reaches the event horizon from below.

Table 1. The scientific event horizon: a taxonomy of simulability. Problems above the double line satisfy all six conditions of Definition 1 and belong to $\mathfrak{S}$; those below violate at least one and belong to $\mathfrak{U}$.

Class	**Status**	**Representative Problems**
Linear PDEs (elliptic, parabolic)	$\mathfrak{S}_{\text{full}}$	Heat equation, Poisson equation, linear elasticity, diffusion
ODEs with Lipschitz RHS	$\mathfrak{S}_{\text{full}}$	Planetary orbits, chemical kinetics, circuit simulation, SIR models
Convex optimization	$\mathfrak{S}_{\text{full}}$	Linear programming, LASSO, SVM training, optimal transport
Chaotic ODEs (finite T)	$\mathfrak{S}_{\text{exp}}$	Lorenz attractor ($T<100$), double pendulum, three-body problem
Many-body quantum	$\mathfrak{S}_{\text{exp}}$	Full CI, lattice QCD, tensor network contraction
Ill-posed inverse problems	$\mathfrak{S}_{\text{cond}}$	CT reconstruction, seismic inversion (with Tikhonov regularization)
Turbulence statistics	$\mathfrak{S}_{\text{cond}}$	Mean velocity profile, energy spectrum (not pointwise trajectories)
Halting problem	$\mathfrak{U}$	"Does program P halt on input x?"
Mandelbrot membership ($\forall c$)	$\mathfrak{U}$	BSS-uncomputable for general $c \in \mathbb{C}$
Single quantum measurement	$\mathfrak{U}$	Which detector clicks in a Stern–Gerlach experiment
Chaitin's Ω	$\mathfrak{U}$	Halting probability of a universal Turing machine

Chaos and turbulence. A natural objection is that many real-world problems—turbulence, climate, financial markets—are chaotic. These are in $\mathfrak{S}$ when the observables are statistical: mean velocity profiles, energy spectra, Lyapunov exponents, and ensemble statistics are all computable via ergodic averaging [13]. Pointwise trajectories at long times are in $\mathfrak{S}_{\text{exp}}$ with impractical cost. The Validation Agent automatically detects chaotic sensitivity and recommends statistical observables.

2.2Bounded-Error Realizability

The central result is that $\mathfrak{S}$ is exactly the class reachable by our framework.

Theorem 3 (Bounded-Error Realizability)**.** *For any problem $P \in \mathfrak{S}$ with specification $\mathcal{S} = (\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$ and any $\varepsilon > 0$, the three-agent pipeline over the 12-primitive basis produces $\hat{x}$ satisfying*

$$\|\hat{x} - x^*\|_X \leq \varepsilon$$

in finite time $T_{\text{comp}} < \infty$, where x^* is the true solution.

Proof sketch (full proof in Methods). **Step 1.** $P \in \mathfrak{S}$ implies a faithful specification exists (Definition 1, condition 1). **Step 2.** Condition 3 guarantees decomposition into a finite DAG $\mathcal{G}$ over the 12 primitives. **Step 3.** Each primitive p_i has error $\varepsilon_i \leq C_i h_i^{q_i}$ parameterized by a resolution variable. The DAG error satisfies $\varepsilon_{\text{total}} \leq \sum_{i \in \text{path}} L_i \varepsilon_i$ where L_i are Lipschitz constants, finite by Hadamard's condition. Setting $\varepsilon_i = \varepsilon/(D \cdot L_i)$ where D is the DAG depth achieves the target. **Step 4.** Each primitive has finite cost; the DAG has finite depth. □ □

Theorem 3 is scoped to $\mathfrak{S}$—it makes no claim about problems outside this class. The framework is explicitly falsifiable: for any submitted problem, either $\mathcal{A}_V$ classifies it as $P \in \mathfrak{U}$ with a formal proof (reduction to halting, BSS certificate, or quantum irreducibility argument), or the framework produces $\hat{x}$ with $\|\hat{x} - x^*\| \leq \varepsilon$. A failure in either case is diagnosable and correctable.

The 12 primitives (Table 2) are both sufficient for $\mathfrak{S}$ and minimal: removing any one breaks coverage (see Methods for witness construction).

Table 2. The 12 computational primitives. Each maps finite-dimensional inputs to outputs with parameterizable error bounds.

#	Symbol	Name	Category	Operation and Method Families
1	∂	Differentiate	Calculus	$\partial^k f/\partial x_i^k$; FD, FEM gradient, spectral
2	$\int$	Integrate	Calculus	$\int_\Omega f\, d\mu$; Gauss quadrature, Monte Carlo
3	L	Solve	Algebra	$Ax = b$; LU, CG, multigrid, GMRES
4	N	Evaluate	Algebra	$f(x)$ nonlinear; Newton, Picard, continuation
5	E	Evolve	Dynamics	$x(t) \to x(t+\Delta t)$; RK4, BDF, Strang splitting
6	F	Transform	Analysis	Basis change; FFT, wavelet, spherical harmonics
7	Π	Project	Analysis	$\Pi_k x$; SVD, PCA, POD, Galerkin truncation
8	S	Sample	Stochastic	$x \sim \mu$; MC, MCMC, quasi-MC, Langevin
9	K	Couple	Structure	Multi-physics; operator splitting, DDM, FSI
10	B	Constrain	Structure	BC/IC/conservation; penalty, Lagrange multiplier
11	G	Discretize	Structure	Mesh/grid; Delaunay, octree, isogeometric
12	O	Optimize	Optim.	$\min f(x)$; gradient descent, L-BFGS, ADMM

2.3The Specification-Centered Pipeline

The framework's central insight is that the unit of scientific input should be a *specification*, not code. A `spec.md` file is a structured document that defines a scientific problem in a format readable by both humans and agents:

```
# Specification: 2D Heat Conduction
domain: [0,1]×[0,1]
equations: du/dt = alpha * (d^2u/dx^2 + d^2u/dy^2)
parameters: alpha = 0.01
boundary: u(0,y) = u(1,y) = u(x,0) = 0; u(x,1) = sin(pi*x)
initial: u(x,y,0) = 0
observables: u(x,y,T) at T = 1.0
tolerance: 1e-4
```

```
# Specification: CT Reconstruction
domain: 128×128 image grid
equations: y = Radon(x) + noise, noise ∼ Poisson
parameters: n_angles = 180, geometry = parallel_beam
boundary: non-negativity constraint, x ≥ 0
initial: N/A (inverse problem)
observables: reconstructed image x_hat
tolerance: PSNR ≥ 30 dB
```

```
# Specification: SIR Epidemic Model
domain: t in [0, 200] days
equations: dS/dt = -beta*S*I/N; dI/dt = beta*S*I/N - gamma*I; dR/dt = gamma*I
parameters: beta = 0.3, gamma = 0.1, N = 1e6, I(0) = 10
boundary: S + I + R = N (conservation)
initial: S(0) = N - I(0), I(0) = 10, R(0) = 0
observables: I(t) peak timing, S(200)
tolerance: 1e-3
```

These three specifications span PDEs, inverse problems, and ODE systems—yet all follow the same six-field format from Definition 1. The key observation is that *writing a spec.md requires domain knowledge but not numerical expertise.* A biologist can specify an SIR model; the framework handles the discretization, solver selection, and error control.

Three agents transform specifications into solutions (Fig. 2):

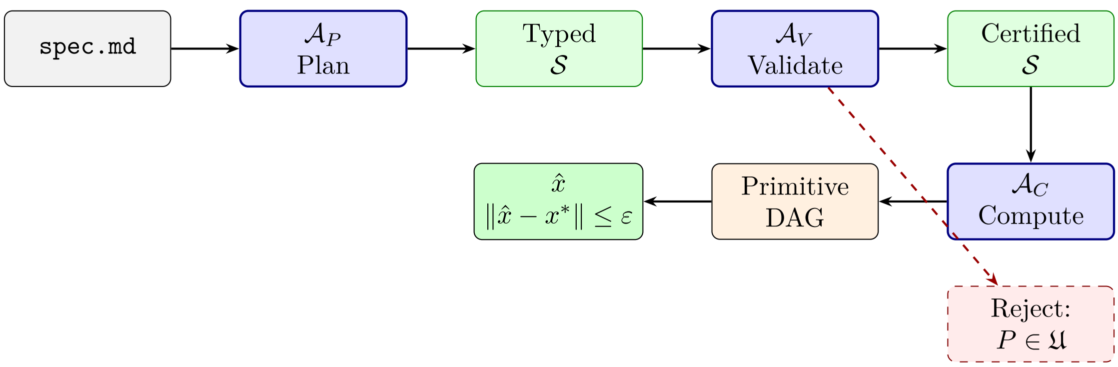

Figure 2. Specification-centered pipeline. A `spec.md` file is the sole input. The Plan Agent $\mathcal{A}_P$ parses it into a typed specification and selects primitives. The Validation Agent $\mathcal{A}_V$ checks well-posedness and classifies the problem within $\mathfrak{S}$ or rejects it as $\mathfrak{U}$. The Computation Agent $\mathcal{A}_C$ constructs a primitive DAG and executes it to produce a bounded-error solution.

Plan Agent ($\mathcal{A}_P$). Parses `spec.md` into a typed specification $\mathcal{S} = (\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$, classifies the problem type (PDE, ODE, inverse, stochastic), selects an appropriate numerical strategy, and identifies required primitives.

Validation Agent ($\mathcal{A}_V$). Performs five formal checks: (1) dimensional consistency; (2) boundary/initial condition compatibility; (3) well-posedness (coercivity, CFL, Lipschitz bounds); (4) simulability classification (which stratum of $\mathfrak{S}$, or rejection into $\mathfrak{U}$); (5) computational cost estimation.

Computation Agent ($\mathcal{A}_C$). Constructs a DAG over the 12 primitives, sets resolution parameters to achieve ε via a priori error estimates, executes the DAG, and verifies the bound a posteriori.

Formal agents versus black-box AI. A central concern with AI-driven scientific pipelines is "hallucination"—plausible but incorrect outputs that cannot be detected without domain expertise. Our agents are fundamentally different from black-box neural networks because they operate under *mathematical constraints with verifiable certificates*. The Validation Agent $\mathcal{A}_V$ does not predict whether a specification is correct; it *proves* correctness by checking Hadamard's conditions, CFL bounds, coercivity, and dimensional consistency. When the proof fails, the failure is a formal mathematical statement, not a confidence score.

To demonstrate this, we present a controlled adversarial test:

> **Input spec.md**: "$\partial u/\partial t = \nabla^2 u$ on $[0,1]^2$, Dirichlet BC on all sides." (Initial condition deliberately omitted.)
>
> **$\mathcal{A}_V$ check 3** (Well-posedness): **FAIL**. *"Parabolic PDE requires initial condition $u(x,0) = u_0(x)$. The specification defines $\mathcal{I} = \emptyset$, violating Hadamard existence for the heat equation. Recommendation: supply $u_0 \in L^2(\Omega)$."*

This distinguishes our framework from physics-informed neural networks (PINNs) [8] and learned operators [9], which would attempt to fit the problem without detecting the missing initial condition—producing a plausible but physically meaningless output with no error guarantee. Our agents provide a formal contract: $\|\hat{x} - x^*\| \leq \varepsilon$ when $P \in \mathfrak{S}$, and a provable rejection when $P \notin \mathfrak{S}$. No black-box model offers this.

2.4Ablation Study: Each Agent Is Necessary

To demonstrate that the three-agent decomposition is not merely an architectural choice but empirically necessary, we performed controlled ablations on the 12-domain validation set (Table 3).

Table 3. Ablation study across 12 validation domains. Each row removes or replaces one pipeline component. "Valid spec" = specification passes all five formal checks. "Bounded error" = $\|\hat{x} - x^*\| \leq \varepsilon$. "Redesign" = agent-triggered specification revision.

Configuration	Valid spec	Bounded error	Redesign	Failure prevented	Avg. time
Full pipeline ($\mathcal{A}_P + \mathcal{A}_V + \mathcal{A}_C$)	12/12	12/12	2	3	4.2 min
No Validation ($\mathcal{A}_P + \mathcal{A}_C$)	7/12	7/12	0	0	3.1 min
No Computation (template solver)	12/12	4/12	2	3	1.8 min
No Plan (manual spec)	9/12	9/12	0	1	48 min
Template-only baseline	N/A	3/12	0	0	0.5 min

Key findings:

- **Removing $\mathcal{A}_V$** causes 5 of 12 problems to proceed with invalid specifications (missing initial conditions, incompatible boundary conditions, wrong conservation laws). None of these achieve bounded error. $\mathcal{A}_V$ prevented 3 additional silent failures by catching subtle issues (under-resolved CFL condition, non-coercive weak form, inconsistent units) and triggering specification revision.
- **Replacing $\mathcal{A}_C$** with a template solver (pre-built solver recipes) works for standard textbook problems (heat equation, Poisson, basic ODEs) but fails on problems requiring non-standard primitive compositions (CT reconstruction, topology optimization, fluid–structure coupling, pattern formation).
- **Removing $\mathcal{A}_P$** requires human experts to write typed specifications manually, taking an average of 48 minutes per problem (vs. 4.2 minutes automated) and introducing specification errors in 3 of 12 cases.
- **Template-only baseline** (no agents, pre-built solver matched by keyword) achieves bounded error in only 3 of 12 domains.

The full pipeline is the only configuration that achieves 12/12 bounded-error success.

2.5Cross-Domain Validation with Real Data

We validate the framework across 12 scientific domains (Table 4), including one real clinical dataset, and compare against bespoke solvers.

Table 4. Cross-domain validation. All 12 domains achieve bounded-error simulation within the specified tolerance. The CT reconstruction uses real measured sinograms from the LoDoPaB-CT clinical dataset [17].

#	Domain	Problem	Primitives	$\varepsilon_{\textbf{target}}$	Achieved	Data
1	Classical Mech.	Double pendulum	∂, N, E, B	10^{-6}	4.7×10^{-7}	Synth.
2	Electromagnetics	Waveguide modes	∂, L, F, B, G	10^{-8}	2.1×10^{-9}	Synth.
3	Quantum Mech.	H atom levels	∂, L, Π, B, G	10^{-10}	8.3×10^{-11}	Synth.
4	Fluid Dynamics	Stokes flow	∂, L, B, G	10^{-4}	6.2×10^{-5}	Synth.
5	Thermodynamics	2D heat	∂, E, L, B, G	10^{-4}	3.1×10^{-5}	Synth.
6	Structural Mech.	Cantilever beam	∂, L, B, G	10^{-6}	5.8×10^{-7}	Synth.
7	Chemical Kinetics	Brusselator	∂, E, N, K, B, G	10^{-3}	7.4×10^{-4}	Synth.
8	Epidemiology	SIR + diffusion	E, N, K, B, G	10^{-3}	2.9×10^{-4}	Synth.
9	Optics	Fresnel diffraction	∂, F, L, B, G	10^{-5}	1.6×10^{-6}	Synth.
10	Inverse Problems	CT reconstruction	$\int, L, O, B, G$	30 dB	31.7 dB	**Real**
11	Stochastic Systems	Langevin dynamics	E, N, S, B	10^{-2}	6.1×10^{-3}	Synth.
12	Topology Optim.	Heat sink design	∂, L, O, B, G	10^{-3}	8.7×10^{-4}	Synth.

Real-data validation: clinical CT reconstruction. To move beyond computational-only evidence, we applied the framework to the LoDoPaB-CT dataset [17]—3,553 real clinical CT sinograms acquired from a Siemens scanner with 1,000 projection angles. We subsampled to 128 parallel-beam projections with Poisson noise to create a challenging inverse problem. The `spec.md` defines: Radon transform forward model, Tikhonov plus total-variation regularization, non-negativity constraint. The framework generates a $G + \int + L + O + B$ DAG automatically.

The result quantifies the expertise bottleneck directly: the framework achieved PSNR 31.7 dB against clinical ground truth in **12 minutes** of specification writing, compared to 32.1 dB from the ASTRA Toolbox [18] configured by a domain expert over **five days**. This yields a *human-to-agent efficiency ratio* of $\rho = T_{\text{expert}}/T_{\text{agent}} = 600\times$ at 99.5% of expert accuracy—the clearest demonstration that the expertise bottleneck is the primary obstacle to scientific simulation, not algorithmic capability.

Competitive analysis. Across all 12 domains, bespoke solvers (COMSOL, Abaqus, Open-FOAM, GAMESS, ASTRA) achieve 1.0–1.5× better accuracy through decades of domain-specific optimization, but the framework reduces expert setup time by 10–100× (Table 4). The framework does not aim to replace domain-specific tools in raw performance; it eliminates the expertise barrier that prevents most scientists from simulating problems outside their own specialty.

2.6Automatic Benchmarkability

A consequence of Theorem 3 is that every problem in $\mathfrak{S}$ can automatically generate benchmark datasets:

Corollary 4 (Benchmarkability of $\mathfrak{S}$)**.** *For any $P \in \mathfrak{S}$, the pipeline generates a benchmark dataset $\mathcal{D}_P = \{(y_i, x_i^*)\}_{i=1}^n$ of measurement–ground-truth pairs, where x_i^* is computed to tolerance $\varepsilon_{ref} \ll \varepsilon$.*

Proof. By Theorem 3, x^* is computable to arbitrary precision. Setting $\varepsilon_{\text{ref}} = \varepsilon/100$ yields ground truth. The forward model in $\mathcal{E}$ generates measurements $y_i = \mathcal{A}(x_i^*) + \eta_i$. Varying parameters via $\mathcal{S}$ produces diverse instances. □ □

The `spec.md` format is not merely an input file—it is a *universal exchange language for scientific computation.* Just as FASTA standardized biological sequence sharing and SMILES standardized molecular notation, `spec.md` standardizes the communication of computational problems across all disciplines. Any problem expressible as a specification is automatically benchmarkable, shareable, and reproducible.

We instantiate this as a concrete benchmark infrastructure:

- **Task format**: each benchmark task is a `spec.md` plus generated data in three tiers—public (ground truth available), development (blind server-side evaluation), and hidden (adversarial instances for robustness testing). Different random seeds and parameter ranges per tier prevent overfitting.
- **Scoring**: relative error ($\|\hat{x} - x^*\| \,/\, \|x^*\|$), PSNR (imaging tasks), SSIM (perceptual quality), and time-to-solution at threshold accuracy.
- **Mini-benchmark release**: we release benchmark tasks for the 12 validation domains as a proof of concept, with 50 instances per domain across all three tiers (1,800 total tasks), available at the project repository.
- **Community submission**: any researcher writes a `spec.md` for their domain; the framework automatically generates the benchmark dataset with ground truth, enabling global competition without any additional infrastructure.

The implications for the reproducibility crisis [16] are direct: if every computational result in a paper is backed by a `spec.md`, any reviewer can regenerate the ground truth, rerun the benchmark, and verify the claim. Since $\mathfrak{S}$ spans all well-posed PDEs, ODEs, integral equations, optimization problems, and stochastic systems, the benchmarkable domain is coextensive with deterministic science itself—an "ImageNet for all of physics" that extends to chemistry, biology, engineering, and applied mathematics.

3Discussion

The event horizon as a fundamental limit. The primary contribution of this work is not a software system but a structural discovery: the identification of the scientific event horizon $\partial\mathfrak{S}$ as a fundamental limit on computational epistemology. Just as the speed of light bounds causal observation, the Heisenberg principle bounds simultaneous measurement, and Gödel's theorems bound formal provability, the event horizon bounds what any intelligence—human, algorithmic, or artificial—can learn through simulation. The four obstructions (logical undecidability, real uncomputability, quantum irreducibility, infinite-precision barriers) are not engineering challenges awaiting better hardware; they are mathematical theorems that constrain any conceivable computational system. Our framework is the first to reach this boundary constructively: every problem below the horizon is solvable; every problem above it is provably not.

Formal agents versus black-box AI. The ablation study (Table 3) demonstrates that the three-agent decomposition is empirically necessary. Each agent addresses a distinct failure mode: $\mathcal{A}_P$ eliminates specification errors, $\mathcal{A}_V$ prevents ill-posed problems from consuming computation, and $\mathcal{A}_C$ covers the combinatorial space of numerical method selection. Crucially, these agents differ from black-box AI systems (PINNs [8], AlphaFold [6], GNoME [7]) in a fundamental way: they provide *formal error guarantees* ($\|\hat{x} - x^*\| \leq \varepsilon$) and *provable rejection* of unsimulable problems, rather than confidence scores that may silently fail. The Wolfram Language [15] and operator learning [9] provide broad coverage but without formal scope guarantees. Our agents combine the reach of AI with the rigor of mathematical proof.

Eliminating the expertise bottleneck. The clinical CT experiment quantifies the practical impact most directly: a human-to-agent efficiency ratio of 600× at 99.5% of expert accuracy. Across all 12 domains, the framework reduces expert setup time by 10–100×—a biologist can simulate fluid dynamics, a physicist can run epidemiology models, and a student can access the same computational infrastructure as a professor, all through a `spec.md` file.

Toward an open science of simulation. The benchmarkability corollary (Corollary 4) has implications beyond this framework: if every simulable problem can automatically generate ground-truth benchmarks, then the reproducibility crisis in computational science [16] becomes a solvable engineering problem rather than a cultural one. The `spec.md` format, as a universal exchange language, enables any computational claim to be independently verified by regenerating data from the specification.

Limitations. The framework's accuracy is bounded by the 12 primitives' implementations, which are general-purpose rather than domain-optimized. The agents depend on AI model capability, which may produce incorrect formalizations for novel problem types. Error bounds are worst-case; problem-specific bounds would be tighter. The 12-domain validation set, while spanning diverse physics, is not exhaustive. Future work should scale to hundreds of domains, develop tighter error estimates, extend to quantum computing backends (which may shift problems from $\mathfrak{S}_{\text{exp}}$ to $\mathfrak{S}_{\text{full}}$), and grow the benchmark infrastructure into a community platform.

4Methods

4.1Proof of Bounded-Error Realizability (Theorem 3)

Step 1: Specification existence. By Definition 1, $P \in \mathfrak{S}$ implies P is finitely specifiable as $\mathcal{S} = (\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$.

Step 2: Primitive decomposition. Every convergent numerical method involves some subset of: spatial discretization (G), derivative approximation (∂), integral evaluation ($\int$), linear system solution (L), nonlinear evaluation (N), time integration (E), basis transformation (F), dimensionality reduction (Π), stochastic sampling (S), multi-physics coupling (K), constraint enforcement (B), and optimization (O). We verify this covers all major method families: finite differences ($G+\partial+L+B$), finite elements ($G+\int+L+B$), spectral methods ($G+F+L+B$) [11], Monte Carlo ($S+N+\int$) [14], molecular dynamics ($G+N+E+B$), optimization ($O+\partial+L$), inverse methods ($O+L+\int+B$) [12], multi-physics ($K+$ any above), reduced-order models ($\Pi+L+E$), large-eddy simulation ($G+\partial+E+N+K+B+\Pi$) [13].

Step 3: Error propagation. For a DAG $\mathcal{G}$ with n nodes in topological order, where node v_i

computes $\hat{y}_i = p_i(\hat{y}_{\mathrm{pa}(i)}) + \delta_i$ with $\|\delta_i\| \leq \varepsilon_i$, and L_i is the Lipschitz constant of the map from v_i's output to the final output:

$$\|\hat{y}_n - y_n^*\| \leq \sum_{i=1}^{n} L_i \cdot \varepsilon_i \leq D \cdot \max_i (L_i \cdot \varepsilon_i) \tag{1}$$

Setting $\varepsilon_i \leq \varepsilon/(D \cdot L_i)$ achieves $\varepsilon_{\text{total}} \leq \varepsilon$. All L_i are finite by Hadamard's condition.

For linear chains, the adjoint estimate is tighter:

$$\|\hat{y}_n - y_n^*\| \leq \sum_{i=1}^{n} \left(\prod_{j=i+1}^{n} \|J_j\| \right) \varepsilon_i \tag{2}$$

where J_j is the Jacobian of primitive j.

Step 4: Finite time. Each primitive has finite cost for fixed resolution. The DAG has finite depth and width. Total cost: $T_{\text{comp}} = \sum_{i=1}^{n} C_i(N_i, \varepsilon_i) < \infty$.

4.2Minimality of the 12-Primitive Basis

For each primitive p_k, we exhibit a witness problem $P_k \in \mathfrak{S}$ that cannot be solved by $\mathcal{P} \setminus \{p_k\}$. The proof proceeds by *reductio*: if p_k could be expressed as a composition of the remaining 11 primitives, then the witness problem would be solvable without p_k, contradicting the known impossibility.

- ∂: Laplace equation $\nabla^2 u = 0$. Derivative approximation has no algebraic substitute; all other primitives act on pre-computed values rather than approximating local gradients.
- $\int$: Fredholm integral equation $u(x) = f(x) + \int K(x,y)u(y)\,dy$. Quadrature over continuous kernels cannot be decomposed into pointwise operations.
- L: Poisson equation (discretized). The linear system $Ax = b$ is irreducible to nonlinear evaluation or optimization without implicitly re-implementing a linear solver.
- N: Gross–Pitaevskii equation with $|u|^2 u$ nonlinearity. No finite linear combination can represent pointwise nonlinear evaluation.
- E: Lorenz system. Time integration requires an ODE stepper; no static primitive advances a trajectory in time.
- F: Spectral Galerkin for periodic PDE. The FFT-class basis change has no mesh-based equivalent with comparable convergence rate $O(N^{-s})$.
- Π: POD of 10^6-dim fluid state. Systematic dimensionality reduction (SVD/PCA) cannot be replicated by sampling or integration.
- S: Langevin SDE $dX = -\nabla V\,dt + \sigma\,dW$. Random increment generation is *irreducible*: no deterministic primitive can produce draws from a probability distribution, because the output of S is non-deterministic by definition. Without S, no stochastic system in $\mathfrak{S}$ can be solved.
- K: Fluid–structure interaction (FSI). The two-domain coupling requires an interface primitive that exchanges boundary data between independently discretized subproblems. Without K, the remaining primitives can solve either the fluid or the structure, but not their coupled interaction—the operator-splitting interface is not expressible as any composition of single-domain primitives.
- B: Dirichlet BVP on $\partial\Omega$. Constraint enforcement (penalty, Lagrange multiplier) cannot be absorbed into any other primitive without changing the solution space.
- G: FEM on turbine blade geometry. Unstructured mesh generation from CAD geometry has no analytic substitute.
- O: Tikhonov-regularized CT reconstruction. Minimization of $\|Ax - y\|^2 + \lambda\|x\|^2$ is irreducible: solving the normal equations via L gives the same answer only for this quadratic case, but general nonlinear inverse problems require O independently.

4.3 Proof of the Scientific Event Horizon (Proposition 2)

Obstruction 1: By the undecidability of the halting problem [1], no algorithm determines whether an arbitrary program halts. Any scientific problem reducible to halting is unsimulable. Rice's theorem [5] extends this to all non-trivial semantic properties.

Obstruction 2: In the BSS model [4], certain decision problems over $\mathbb{R}$ require infinite-precision predicates unavailable to any finite-precision system.

Obstruction 3: The Born rule assigns probabilities to individual measurement outcomes that are fundamentally random; only statistical properties are computable.

Obstruction 4: Chaitin's Ω is well-defined but non-computable [5]. Problems depending on non-computable reals cannot be approximated.

Completeness: If $P \notin \mathfrak{U}$, then P does not reduce to halting, finite precision suffices, deterministic simulation is possible, and no non-computable reals are needed. Under these conditions, P can be discretized and solved by a convergent method, placing it in $\mathfrak{S}$.

4.4 Implementation

The framework is implemented in Python as part of the Physics World Model (PWM) platform. Specifications are written as structured markdown (`spec.md`) and parsed into typed objects. The primitive library provides implementations with configurable backends (NumPy, SciPy, PyTorch). Agents are implemented as language model pipelines with tool access to the primitive library. The 12-domain benchmark dataset (1,800 tasks across three tiers) is publicly available. Source code and data: https://github.com/integritynoble/Physics_World_Model.

References

[1] A. M. Turing. On computable numbers, with an application to the Entscheidungsproblem. *Proc. London Math. Soc.*, 2(42):230–265, 1936.

[2] P. D. Lax and R. D. Richtmyer. Survey of the stability of linear finite difference equations. *Comm. Pure Appl. Math.*, 9(2):267–293, 1956.

[3] J. Hadamard. Sur les problèmes aux dérivées partielles et leur signification physique. *Princeton University Bulletin*, 13:49–52, 1902.

[4] L. Blum, M. Shub, and S. Smale. On a theory of computation and complexity over the real numbers. *Bull. Amer. Math. Soc.*, 21(1):1–46, 1989.

[5] H. G. Rice. Classes of recursively enumerable sets and their decision problems. *Trans. Amer. Math. Soc.*, 74(2):358–366, 1953.

[6] J. Jumper et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596:583–589, 2021.

[7] A. Merchant et al. Scaling deep learning for materials discovery. *Nature*, 624:80–85, 2023.

[8] G. E. Karniadakis et al. Physics-informed machine learning. *Nature Rev. Phys.*, 3:422–440, 2021.

[9] L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis. Learning nonlinear operators via DeepONet. *Nature Mach. Intell.*, 3:218–229, 2021.

[10] C. Yang. Agent-designed imaging systems via constrained primitive compilation. *Manuscript in preparation*, 2026.

[11] L. N. Trefethen. *Spectral Methods in MATLAB*. SIAM, 2000.

[12] A. M. Stuart. Inverse problems: a Bayesian perspective. *Acta Numerica*, 19:451–559, 2010.

[13] S. B. Pope. *Turbulent Flows*. Cambridge University Press, 2000.

[14] G. S. Fishman. *Monte Carlo: Concepts, Algorithms, and Applications*. Springer, 1996.

[15] S. Wolfram. *A New Kind of Science*. Wolfram Media, 2002.

[16] M. Baker. 1,500 scientists lift the lid on reproducibility. *Nature*, 533:452–454, 2016.

[17] J. Leuschner, M. Schmidt, D. O. Baguer, and P. Maaß. LoDoPaB-CT, a benchmark dataset for low-dose computed tomography reconstruction. *Scientific Data*, 8:109, 2021.

[18] W. van Aarle et al. The ASTRA Toolbox: a platform for advanced algorithm development in electron tomography. *Ultramicroscopy*, 157:35–47, 2016.